Diagnosis of Sport Injuries with Machine Learning: Explanation of Induced Decisions

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Abstract

Machine learning techniques can be used to extract knowledge from data stored in medical databases. In our application, various machine learning algorithms were used to extract diagnostic knowledge to support decisions in the diagnosis of sport injuries.

1 Introduction

Machine learning technology is well suited for the induction of diagnostic and prognostic rules and for solving specialized diagnostic and prognostic problems. Data about correct diagnoses are often available in archives of specialized hospitals and clinics, where the number of stored cases grows daily. Similar data gathering is done also in daily routine of specialist medical doctors.

Such data gathering occurs also in the Center for Sport Medicine of the Ljubljana University Medical Hospital, where records of patients with sport injuries are collected daily. This work is limited to data analysis of patient records with injuries in athletics and handball. The aim of our work is to provide systematic computer-supported data gathering and storing, intelligent analysis of stored data, support of diagnostic decisions, and the transfer of expert diagnostic knowledge from the experienced specialist to young nonexperienced medical doctors. An important aspect, which has motivated this study, is to reveal unclear influences of individual anamnestic and clinical parameters for individual diagnoses. Moreover, to support diagnostic decisions, reasonably high diagnostic accuracy has to be achieved, as well as the transparency (explanation) of proposed decisions.

In recent years, many different machine learning systems were developed. Machine learning methods can be classified into three major groups: inductive learning of symbolic rules (such as induction of if-then rules, decision trees or logic programs), statistical or pattern-recognition methods (such as k-nearest neighbors or instance-based learning, discriminate analysis and Bayesian classifiers), and artificial neural networks (such as networks with backpropagation learning, Kohonen's self-organizing network and Hopfield's associative memory). In this work we are biased towards systems that provide for the explanation

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of proposed decisions, therefore the application of the 'black-box' neural networks was considered inappropriate; we have also limited the selection of systems to several variants of top-down decision tree learners and to several variants of the Bayesian classifier that have proved to be well suited for supporting diagnostic decision making in numerous medical domains [Kononenko, 1993].

The paper presents the problem of sport injury diagnosis, and describes the experiments and results together with the evaluation of the results by a medical expert. One of the important issues addressed in this paper is how to transparently present the induced knowledge to physicians. In order to enhance the explanation capability of the system, we developed a generic expert system shell (incorporating variants of the Assistant algorithm and the Bayesian classifier) which provides several methods enabling a visual representation of the induced knowledge: on-line browsing of decision trees, tables, and pie-charts of various sorts of information, including information gains of attributes for a proposed diagnosis.

2 Diagnostic problem

In the Center for Sport Medicine of the Ljubljana University Medical Hospital, records of patients with sport injuries are collected daily. A patient's first visit to the Center results in a diagnosis of the injury and a treatment recommendation. Patients are usually treated by a series of therapeutic measures in a number of consecutive visits to the Center, as well as with recommended home treatment and exercising.

The current database of athletic and handball injuries consists of 118 patient records, described by the values of 49 attributes. In expert's opinion, the most important diagnostic attributes include, for instance, the localization of injury, the test of forced movement, the Lahman's test, etc., although it is clear that for various diagnoses attributes have a varied diagnostic importance. Diagnoses are grouped into 30 diagnostic classes (the original database deals with more than 50 diagnoses). The most frequent diagnosis is the injury of ligamentary insertions (16% of patients have this diagnosis), thus the majority class is not significantly larger than other classes, which have 11% (injury of muscles of the back side of the tigh), 10% (injury of ankle joint), etc. However, there are 11 diagnostic classes represented by a single training instance, and 4 classes with two training instances each. A reasonable grouping of similar diagnoses only partly solved the problem of classes with too few instances. For example, diagnoses 'distensions of muscles semitendinosus' and 'distension of biceps feminoris' were merged into a common diagnostic group 'injury of muscles of the back side of the tigh' - this is justified by the same location of muscles as they are both located at the back side of the thigh, the same symptoms, similar treatment, and the same physiological cause of the two injuries. For the diagnoses represented by too few instances, expert-defined rules in the form similar to training examples themselves were used in two ways: as pre-classifiers or as generators of additional training instances.

3 Machine learning systems and their explanation capability

In medical diagnosis it is crucial that the system is able to explain and argue about its decisions when diagnosing a new patient. Especially when faced with an unexpected solution of a new problem, the user requires substantial argumentation and explanation.

In this study we used only systems that provide for the explanation of decisions. We used several decision tree learners and several variants of the Bayesian classifier.

Decision tree learners. Decision tree learners include three variants of the Assistant algorithm (Assistant-R, Assistant-I, Assistant-R2) that implement several extensions of the original Assistant algorithm for top-down induction of decision trees [Cestnik, et al., 1987]. The main difference between the algorithms lies in their heuristics for attribute selection: Assistant-I uses informativity whereas Assistant-R and Assistant-R2 use the algorithm ReliefF [Kononenko and Šimec, 1995]. Assistant-R2 is a variant of Assistant-R that generates separate decision trees for each class (diagnosis), and combines these into a classifier for the entire domain. This is in contrast with Assistant-I and Assistant-R that build one general decision tree for the whole domain.

Decision tree learners are known to often give an appropriate explanation: induced decision trees are fairly easy to understand and can be used to support diagnosing without using the computer - this is particularly valuable in situations which require prompt decisions and in situations when computer interaction is psychologically inacceptable. Positions of attributes in the tree, especially the top ones, often directly correspond to domain expert's knowledge. However, in order to produce general rules, these methods use pruning [Cestnik, et al., 1987] which drastically reduces tree sizes. Consequently, the paths from the root to the leaves are shorter, containing only few most informative attributes. Frequently physicians dislike such trees since too few parameters are taken into account and the tree too poorly describes the patients to provide for reliable decisions. Another problem is the variability of decision trees - frequently a small change in the dataset causes a substantial restructuring of the decision trees - this also decreases physician's trust in the proposed diagnosis and in its explanation.

Bayesian classifiers. We used two variants of the Bayesian classifier with extensions for handling continuous attributes: the naive Bayesian classifier and the semi-naive Bayesian classifier. The semi-naive Bayesian classifier is an extension of the naive Bayesian classifier that explicitly searches for dependencies between values of different attributes [Kononenko, 1993]. Both systems require the pre-discretization of continuous attributes. The problem of (strict) pre-discretization is that minor changes in values of the continuous attributes (or, equivalently, minor changes in boundaries) may have a drastic effect on the probability distribution and therefore on the classification. In order to avoid this problem, the naive Bayesian classifier with the fuzzy discretization of continuous attributes was also used [Kononenko, 1993].

Bayesian classifiers induce a table of conditional probabilities which indicate how much a feature (an attribute value) contributes to a diagnosis. When explaining a decision for an individual patient, the explanation of a decision is provided by the indicated 'weigth' of a feature, i.e., the information gain[1] for each patient's feature, as well as the sum of information gains of all features that are in favour or against the decision (diagnosis). One of the main advantages of such a decision, appealing to physicians, is that all the available information is used to explain the decision; such an explanation seems to be 'natural' for medical diagnosis and prognosis.

[1]Information gain (measured in bits) is computed as $-log_2 P(C) + log_2 P(C|V_i)$, where C denotes an individual diagnosis, V_i an individual feature (attribute value), $P(C)$ the prior probability, and $P(C|V_i)$ the conditional probability.

4 Experiments and results

Since the ultimate test of the quality of learners is their performance on unseen cases, experiments were performed on ten different random partitions of the data into 70% training and 30% testing examples. In this way, ten training sets E_i and ten testing sets T_i, $i \in [1..10]$ were generated. In addition, partitions followed the rule that the training set must contain at least half of all examples of each class. In the experiments all the systems used the same training and testing sets. Results of the experiments in terms of the classification accuracy and absolute information score [Kononenko and Bratko, 1991] are outlined below.

Results of experiments using decision tree learners. Table 1 summarizes the results of the Assistant algorithms. All three variants of Assistant achieve approximately the same accuracy and (absolute) information score (note that the accuracy and information score are computed for pruned trees). The comparison of decision trees reveals that Assistant-I selects substantially different attributes than the other two variants and also generates slightly smaller decision trees, which are in turn slightly less accurate.

Classifier	accuracy (%)		inf. score		leaves (#)
	$\overline{x}$	σ	$\overline{x}$	σ	
Assistant-I	58.2	5.8	2.19	0.28	20.9
Assistant-R	62.9	5.7	2.25	0.21	26.3
Assistant-R2	61.7	6.2	2.22	0.06	3.2

Table 1. The performance of the Assistant algorithms, all using the same parameter setting: $m = 2$, prepruning = off, postpruning = on. The number of leaves for Assistant-R2 is the average over 30 trees.

Results of experiments using Bayesian classifiers. As shown in Table 2, the use of fuzzy boundaries significantly improves the classification accuracy of the naive Bayesian classifier. Although the number of continuous attributes is relatively small, strict discretization of continuous attributes exaggerates the importance of those attributes. According to the physician's opinion, the continuous attributes are not very important for classification and fuzzy discretization correctly lowers their influence which drasticaly increases the classification accuracy. The use of the semi-naive Bayesian classifier turns out to be inappropriate for this domain. Joining of values of attributes causes the accuracy to drop. The result suggests that in this domain the attributes are relatively conditionaly independent.

Classifier	accuracy (%)		inf. score	
	$\overline{x}$	σ	$\overline{x}$	σ
naive Bayes - strict	59.4	4.9	1.83	0.15
naive Bayes - fuzzy	69.4	3.0	2.32	0.19
semi-naive Bayes - fuzzy	59.4	4.8	1.82	0.15

Table 2. The performance of the Bayesian classifiers with parameter setting m=2.

5 Physician's evaluation of results

The physician specialist is satisfied with the classification accuracy achieved by the naive Bayesian classifier and estimates this accuracy as acceptable. Besides, he likes the explanation od the decisions provided by of the naive Bayesian classifier: he considers the sum of information gains in favour/against a given diagnosis to be close to the way how physicians diagnose patients. He also prefers the naive Bayesian classifier since it uses all the available attributes for classification.

On the other hand, the decision trees are not considered to be very transparent. In fact the physician specialist feels that the number of attributes in the tree is too small and that the classification with a decision tree ignores significant information about the patient. The decision tree generated by Assistant-I is estimated as non-logical while the decision trees of Assistant-R do replicate the physician specialist's knowledge about the most important attributes and their logical relations.

6 Summary

We developed a generic expert system shell incorporating several machine learning engines that can be used to extract knowledge from data stored in medical databases. The explanation capability of the expert system shell enables the user to get a visual representation of the induced knowledge using several methods of representation: on-line browsing of decision trees, tables, and pie-charts of various sorts of information, including information gains of attributes for a proposed diagnosis.

Experimental results show that the classification accuracy and the explanation capability of the naive Bayesian classifier with the fuzzy discretization of numerical attributes was superior to other methods and was estimated as the most appropriate for practical use in diagnosing of sport injuries.

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